

An Integration Framework Reusing Geochemical Surveys in Crop Nutrient Management

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Abstract—Modern precision agriculture faces a critical Data-Knowledge Gap stemming from the fragmentation of information between high-precision geochemical surveys and agro-ecological requirements and standards. While legacy geological databases contain extensive mappings of stable soil minerals, these resources remain underutilized for real-time nutrient management due to a Feature Asymmetry where predictive soil minerals have minimal overlap with climate-based crop requirements and a Semantic and Scale Mismatch between mass concentration (mg/kg) and application rates (kg/ha). This paper proposes an Information Integration Framework utilizing a Decoupled Pipeline Architecture to bridge these disparate silos. The framework reuses the RHEA Geochemical Dataset ($n = 44,298$) to develop a high-accuracy Virtual Sensor ($R^2 = 0.9516$) that infers Nitrogen (N) status from stable mineral proxies like Aluminum (Al) and Iron (Fe). Simultaneously, a Requirement Estimator trained on an Agro-Ecological Dataset ($n = 2,200$) maps environmental variables (rainfall, temperature) to specific crop needs, demonstrating that non-linear models like KNN ($R^2 = 0.8857$) significantly outperform linear baselines. To resolve unit incompatibility, we implement Relative Percentile Mapping to generate a unit-agnostic Deficiency Score. Our results confirm that integrating chemical science with machine learning enables the reuse of legacy data to provide actionable, crop-specific insights, fostering a more sustainable and information-rich agricultural ecosystem.

Index Terms—Information Integration, Information Reuse, Virtual Sensors, Machine Learning, Data-Knowledge Gap

I. INTRODUCTION

Nutrient management is a cornerstone of modern precision agriculture [1], yet it remains one of the most significant challenges for food security and environmental sustainability. Among essential macronutrients, Nitrogen (N) is the most critical due to its direct impact on crop yield; however, it is also the most volatile and difficult to monitor. Soil nitrogen is the primary limiting nutrient for crop growth in most agricultural soils. Its concentration directly governs crop yield potential, fertilizer prescription rates, and risk of leaching into water bodies [1]. In sub-Saharan Africa specifically, smallholder farmers frequently apply nitrogen without precise soil data, resulting in either chronic under-application [2] or over-application [3] (which drives greenhouse gas emissions, acidification, and eutrophication of freshwater). Traditional methods for determining soil N levels rely on frequent, high-cost laboratory testing, which creates a Data-Knowledge Gap for many agricultural stakeholders. Farmers are often forced

to choose between expensive sampling or relying on “blanket” fertilizer recommendations that lead to poor plant growth.

We propose an Information Integration Framework that bridges two data domains: high-precision geochemical surveys and agro-ecological requirement standards. Stable soil minerals, such as Aluminum (Al), Iron (Fe), and Organic Carbon ($C_{organic}$), are often already mapped in geological databases but rarely utilized for real-time agricultural decision-making. By treating these stable minerals as “proxies,” we can develop virtual sensors to infer current nutrient status without the need for costly chemical analysis. However, integrating these worlds presents significant challenges. First, there is a Semantic and Scale Mismatch; geochemical source data typically measures mass concentration in mg/kg , while agricultural requirements are defined in application rates of kg/ha . Second, Feature Asymmetry exists where the predictive features for soil health (minerals) have zero overlap with the features used to define crop needs (climate and crop type). Finally, Data Sparsity issues, such as the 95% missingness of Phosphorus (P) in primary soil records, complicate data merging.

We benchmark six machine learning algorithms to identify a high-accuracy Virtual Sensor ($R^2 = 0.9516$) and implement a Percentile-based Deficiency Score to create a unit-agnostic Nitrogen Gap analysis. By integrating chemical science with computer science, this framework enables reuse of legacy geological data to provide real-time, crop-specific insights, fostering a more sustainable and information-rich agricultural ecosystem. While the underlying machine learning algorithms are standard, their application to this specific geochemical context provides a previously unattainable capability: the ability to estimate volatile soil nitrogen purely from stable, natively mapped mineral proxies (like Al and Fe). This allows stakeholders to extract continuous, real-time nutrient intelligence from static legacy databases without the need for new, costly chemical soil tests.

II. METHODOLOGY

A. Phase A: The Virtual Sensor Soil Inference Model

The RHEA training dataset [4] contains 44,298 soil observations drawn from sub-Saharan Africa. Our model reuses stable geochemical features, specifically Aluminum (Al), Iron (Fe), Magnesium (Mg), Calcium (Ca), and Organic Carbon ($C_{organic}$) to infer current N status. We benchmark six machine

learning algorithms, including Linear Regression, K-Nearest Neighbors (KNN), Random Forest, and Gradient Boosting variants, to identify the most robust inference engine. XGBoost was selected as the primary model.

B. Phase B: The Requirement Estimator (Crop Model)

A secondary model, the Requirement Estimator, processes the agro-ecological dataset [5]. This model maps environmental variables, such as Rainfall, alongside specific Crop Labels to determine the ideal N level required for optimal yield. By decoupling this from the soil model, the framework remains flexible, allowing integration of new crop requirements without retraining the geochemical inference engine.

C. Phase C: Information Integration Logic

The final stage resolves the semantic and Scale Mismatch between the datasets. The RHEA soil dataset measures Nitrogen in mass concentration (mg/kg), while the requirement dataset uses application rates (kg/ha). To bridge this gap, we implemented Relative Percentile Mapping. Soil Population Ranking, the Actual N (predicted by Phase A), is ranked as a percentile within the global distribution of the geochemical survey. The Crop Population Ranking, the Required N (predicted by Phase B), is similarly ranked within the distribution of agro-ecological requirements. The Deficiency Index is calculated by comparing these two unit-agnostic rankings, identifying the "Nitrogen Gap" regardless of the underlying physical units. The calculation is detailed below:

$$DS = P(X \leq N_{req}) - P(X \leq N_{act}) \quad (1)$$

$P(X \leq N_{req})$ and $P(X \leq N_{act})$ represent the cumulative distribution functions (CDF) of the agro-ecological and geochemical populations, calculated empirically from historical distributions of the RHEA and Agro-Ecological training sets. Consequently, the score represents a relative ranking rather than an absolute physical quantity.

D. Phase D: Data Sparsity and Feature Asymmetry

A primary challenge in this integration was the Feature Asymmetry, where predictive soil features have zero overlap with crop requirement features. Our methodology addresses this by prioritizing Stable Mineral Proxies. By identifying that $C_{organic}$ and other minerals are highly predictive of N , the framework remains robust to incomplete soil tests.

III. RESULTS

A. Dataset Analysis and Comparison

The RHEA training set spans two depth intervals (0–20 cm and 20–50 cm) and a date range 2008–2018. Seventeen geochemical features are recorded per sample. Twelve features, principally the stable minerals (Al, Ca, Fe), macro-nutrients (Mg, K, N), micro-nutrients (Cu, Mn), organic carbon, and pH, are effectively complete, with missing rates below 0.5%. A second cluster of six features (P, B, S, Zn, electrical conductivity, and Na) is simultaneously missing in approximately 95.7% of rows, a pattern consistent with these analytes belonging to

a supplementary analytical suite that was applied to only a subset of sites. As such, we predict Nitrogen with only well-represented features.

A fundamental challenge in merging the RHEA geochemical data with the required climate and crop-management feature sets is the Feature Asymmetry and lack of domain overlap. While agronomic models for Nitrogen (N) typically rely on dynamic variables—such as tillage regimes, fertilizer application rates, and real-time moisture levels—none of these are recorded in the RHEA dataset. Instead, the RHEA data is restricted to stable soil minerals. Our framework bridges this "zero-overlap challenge" by utilizing Aluminum (Al), Iron (Fe), and Organic Carbon ($C_{organic}$) as stable proxies. In these sub-Saharan African soils, Al and Fe indicate clay surfaces that physically protect organic N from decomposition, while $C_{organic}$ serves as a stoichiometric proxy for organic nitrogen pools. The practical consequence is that while our model achieves high predictive accuracy ($R^2 = 0.9516$) by leveraging the soil's state at sampling time, it remains geochemically constrained. It cannot account for human-driven management inputs or transient hydro-physical properties without future integration of remote-sensing or gridded climate products.

TABLE I
RHEA FEATURE AVAILABILITY AND MISSINGNESS SUMMARY.

Feature	Present	Missing	Miss%	Notes
Al	44,296	2	0.0%	Stable mineral; Full coverage
Ca	44,298	0	0.0%	Stable mineral; Full coverage
Fe	44,257	41	0.1%	Stable mineral; Full coverage
C_{org}	44,298	0	0.0%	Organic Carbon; Predictor
N	44,257	41	0.1%	TARGET; Macro-nutrient
pH	44,295	3	0.0%	Soil pH; Master variable
P	1,909	42,389	95.7%	Sparse cluster; Future work
Zn	1,909	42,389	95.7%	Sparse cluster; Future work

TABLE II
GEOMINERAL PROXY CORRELATIONS WITH NITROGEN TARGET.

Proxy Pair	Spearman ρ	Pearson r	Geochemical Relationship
Al vs N	0.545	0.51	Strong (Clay surface protection)
Ca vs N	0.428	0.33	Moderate (pH/aggregate stability)
Fe vs N	0.327	0.26	Weak (Redox sensitivity)

TABLE III
AGRO-ECOLOGICAL REQUIREMENT DATASET PROFILE.

Variable	Mean	Std	Min	Max
Nitrogen (kg/ha)	50.55	36.92	0.00	140.00
pH	6.47	0.77	3.50	9.94
Rainfall (mm)	103.46	54.96	20.21	298.56
Temp. (°C)	25.62	5.06	8.83	43.68

The Requirement Estimator maps environmental variables (Rainfall, Temperature) and Crop Labels to determine ideal N levels. Each dataset is uniquely qualified to support its respective specialized model within the decoupled architecture. The RHEA geochemical dataset is ideal for the Soil Sensor (Phase A) because it provides a high-density, 44,298-observation foundation where stable minerals like Aluminum and Iron exhibit strong, predictable correlations with Nitrogen. These minerals have nearly 100% coverage, allowing the machine learning pipeline, with a variety of model types, to achieve high predictive accuracy ($R^2 = 0.9516$) without being hampered by the "sparse cluster" of secondary nutrients. Conversely, the Agro-Ecological Requirement dataset is perfectly suited for the Requirement Estimator (Phase B) because it offers a clean, 2,200-sample balanced distribution



Fig. 1. Decoupled Pipeline Architecture: Information integration logic bridging geochemical proxies and agro-ecological requirements.

across 22 crop types with zero missing values. This dataset’s inclusion of dynamic environmental factors such as rainfall and temperature allows the model to map specific crop labels to their ideal nitrogen application rates, effectively capturing the non-linear biological needs that stable mineral data alone cannot provide. Together, these datasets allow the framework to reconcile the “Data-Knowledge Gap” by providing precise, domain-specific insights that are later integrated through unit-agnostic percentile mapping.

B. Data-Knowledge Gap: RHEA/Agro-Ecological Datasets

The RHEA Geochemical Dataset and the Agro-Ecological Requirement Dataset lack a common language to provide actionable farming advice. **Feature Asymmetry:** The RHEA dataset contains high-precision soil minerals like **Aluminum (Al)** and **Iron (Fe)**, but it has **zero overlap** with the climate variables such as Rainfall and Temperature found in the Requirement dataset. **Semantic and Scale Mismatch:** RHEA measures Nitrogen in **mass concentration (mg/kg)**, whereas the Requirement dataset defines crop needs in **application rates (kg/ha)**. Consequently, because one cannot mathematically subtract a kg/ha requirement from a mg/kg measurement, the raw data in these two sets cannot be used together to calculate a functional fertilizer recommendation.

C. Solving the Gap through Decoupled Pipeline Architecture

The proposed Decoupled Pipeline Architecture (Figure 1) translates between the two datasets, processing them independently before merging them through a unit-agnostic integration stage. **Soil Inference (Phase A):** The framework reuses the **RHEA dataset** to train a “Virtual Sensor” that transforms stable minerals, which are easily found in legacy geological records, into an accurate estimate of current Nitrogen status. **Requirement Estimation (Phase B):** Simultaneously, it uses the **Requirement dataset** to train a model that maps specific crops, such as Rice or Maize, to their ideal Nitrogen needs based on their local environment. **Integrated Gap Analysis (Phase C):** Instead of attempting direct unit conversion, we rank the outputs from both models as **relative percentiles**. By comparing the soil’s ranking against the crop’s requirement ranking, the system generates a unit-agnostic **Deficiency Score**. This resolves the knowledge gap by providing precise agricultural insights that no longer depend on overlapping features or matching physical units.

D. Modeling

Table IV presents the best 5-fold cross-validated R^2 score achieved by each algorithm after hyperparameter optimization. All six models perform remarkably well, with R^2 scores ranging from 0.9273 (Ridge Regression) to 0.9516 (XGBoost), confirming that the eight geochemical proxy features collectively carry substantial predictive information about soil nitrogen. The top three methods—XGBoost (0.9516), Hist-GradientBoosting (0.9509), and MLP (0.9501)—are tightly clustered at the 0.95 threshold. Ridge Regression, while the weakest performer at 0.9273, still captures the majority of nitrogen variance through linear geochemical relationships, underscoring the strong univariate correlations documented in the dataset analysis. KNN achieves a mid-range score of 0.9378, benefiting from local smoothness but constrained by its inability to extrapolate beyond the training manifold. XGBoost is selected as the champion model based on its superior predictive accuracy and computational efficiency, while the Random Forest (0.9490) is retained as the primary interpretability vehicle due to its native feature importance scores.

E. Feature Importance

The Random Forest feature importance scores (not detailed for brevity) show that Aluminum (Al) emerges as the dominant predictor with an importance score of 0.285, followed by organic carbon (C_{organic}) at 0.230 and calcium (Ca) at 0.175. This ranking is consistent with and strongly supported by the Spearman rank correlations reported in the dataset analysis. pH ranks fourth (0.120), reflecting its known role as a master variable in soil chemistry. Iron (Fe), magnesium (Mg), potassium (K), and manganese (Mn) contribute progressively smaller fractions of predictive power, consistent with their secondary roles as cofactors and redox-sensitive modifiers rather than primary controllers of nitrogen retention.

F. Phase B Results

The Requirement Estimator results (Table V) reveal a significant performance gap between linear and non-linear modeling approaches when mapping agro-ecological variables to nitrogen requirements. As shown in the benchmarking results, non-linear models such as K-Nearest Neighbors (KNN), Random Forest, XGBoost, and LightGBM all achieved robust predictive accuracy, with R^2 scores ranging from 0.877 to 0.886. Notably, KNN emerged as the most efficient and accurate model for this dataset, achieving the highest R^2 of 0.8857. In stark contrast, the linear baseline provided by Ridge Regression failed to capture the underlying relationships, yielding an R^2 of only 0.1148. This disparity validates the necessity of the complex, decoupled architecture proposed in this framework to handle the non-linear biological and climatic factors that determine crop nutrient needs.

G. Decoupled Pipeline Architecture

The Virtual Sensor reuses the RHEA Geochemical Dataset to train an ML regressor that transforms eight stable mineral features, such as Aluminum (Al) and Iron (Fe), into an

TABLE IV
PHASE A BENCHMARKING: SOIL NITROGEN PREDICTION

Algorithm	R^2 Score	Time (s)
XGBoost	0.9516	4.34
HistGradientBoosting	0.9509	12.58
MLP (Tiny)	0.9501	45.87
Random Forest (Memory-Safe)	0.9490	45.39
KNN (Stable)	0.9378	6.08
Ridge Regression	0.9273	0.13

TABLE V
PHASE B: NITROGEN REQUIREMENTS ESTIMATION

Algorithm	Best R^2	Time (s)
K-Nearest Neighbors (KNN)	0.8857	0.52
Random Forest	0.8829	66.73
XGBoost	0.8786	10.07
LightGBM	0.8767	7.46
Multi-Layer Perceptron (MLP)	0.8512	70.46
Ridge Regression	0.1148	2.77

TABLE VI
INTEGRATED GAP ANALYSIS FOR SPECIFIC CROPS.

Crop	Act. N (mg/kg)	Req. N (kg/ha)	Defic. Score
Rice	2.5067	74.81	0.3660
Maize	2.5067	80.97	0.4100
Coffee	2.5067	97.49	0.5280
Lentil	2.5067	41.77	0.1300

estimate of the soil’s actual Nitrogen (N) status. Simultaneously, the Agronomist model is trained on the Agro-Ecological dataset to predict the Nitrogen requirements of specific crops based on environmental variables like rainfall and temperature. The experimental results demonstrate the high predictive accuracy and functional integration of the proposed framework. The final Integrated Gap Analysis effectively reconciled the scale mismatch between the two datasets. As shown in Table VI, raw values from both domains were transformed into a unit-agnostic Deficiency Score. It is important to note that these examples are currently illustrative proofs-of-concept. At this stage, the Deficiency Score serves as a relative heuristic to prioritize at-risk regions or crops, rather than a calibrated tool for specific fertilizer prescription rates.

IV. DISCUSSION

The feature importance hierarchy produced by the Random Forest model is coherent with established soil science theory, lending external validity. Aluminum’s primacy as a nitrogen predictor is well-grounded in pedology: high aluminum concentrations are a hallmark of highly weathered, fine-textured Ferralsols and Acrisols that dominate the humid and sub-humid regions of sub-Saharan Africa. These soils possess large clay surface areas, primarily kaolinite and gibbsite, that physically protect organic matter from microbial decomposition through organo-mineral associations, thereby accumulating both organic carbon and organic nitrogen in proportion to their mineralogical surface area. The model’s high weighting of organic carbon (C_{organic}) is equally well-founded: soil organic matter is the dominant reservoir of plant-available nitrogen in tropical soils, and the tight stoichiometric coupling between C and N in microbial biomass (C:N 8–12) means that C_{organic} is essentially a scaled proxy for organic N pools under

stable land use. Calcium’s importance reflects its role in aggregate stability: calcium ions bridge negatively charged clay and organic matter surfaces, forming stable organo-mineral complexes that sequester nitrogen against mineralisation. pH’s contribution as the fourth-ranked predictor is consistent with its role as a master variable governing microbial nitrogen cycling: nitrogen mineralisation rates peak at pH 6–7, and strongly acidic soils (pH $\leq$ 5.0, common in weathered African soils) suppress nitrification, reducing inorganic N availability and concentrating N in organic forms. Taken together, the model architecture and its learned feature hierarchy reflect the actual biogeochemical mechanisms controlling nitrogen distribution across sub-Saharan African soils, confirming that the benchmark pipeline has produced a scientifically interpretable and practically meaningful nitrogen model rather than a purely empirical black box.

V. CONCLUSION AND FUTURE WORK

The proposed Information Integration Framework successfully bridges the “Data-Knowledge Gap” by reusing legacy geochemical surveys as a high-precision virtual sensor for agricultural nutrient management, achieving a peak predictive accuracy of $R^2 = 0.9516$ for soil nitrogen. By implementing a decoupled architecture and unit-agnostic percentile mapping, this research overcomes significant semantic mismatches and the “zero-overlap challenge” inherent in merging disparate geological and agro-ecological datasets. While the current model provides a robust foundation for identifying nitrogen deficiencies in sub-Saharan African soils using stable mineral proxies like Aluminum and Iron, future work will focus on addressing data sparsity in phosphorus records. While Phosphorus (P) is fully mapped in the agro-ecological requirements, its extreme 95.7% sparsity in the RHEA geochemical records prevents the creation of a reliable Virtual Sensor for this macronutrient. Furthermore, while the Deficiency Score mathematically resolves the unit mismatch, it has not yet been validated against biological yield responses in the field. A critical next step is to conduct empirical field trials to calibrate these relative percentile rankings against actual agronomic outcomes, establishing the score’s rigorous validity for prescriptive decision support.

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